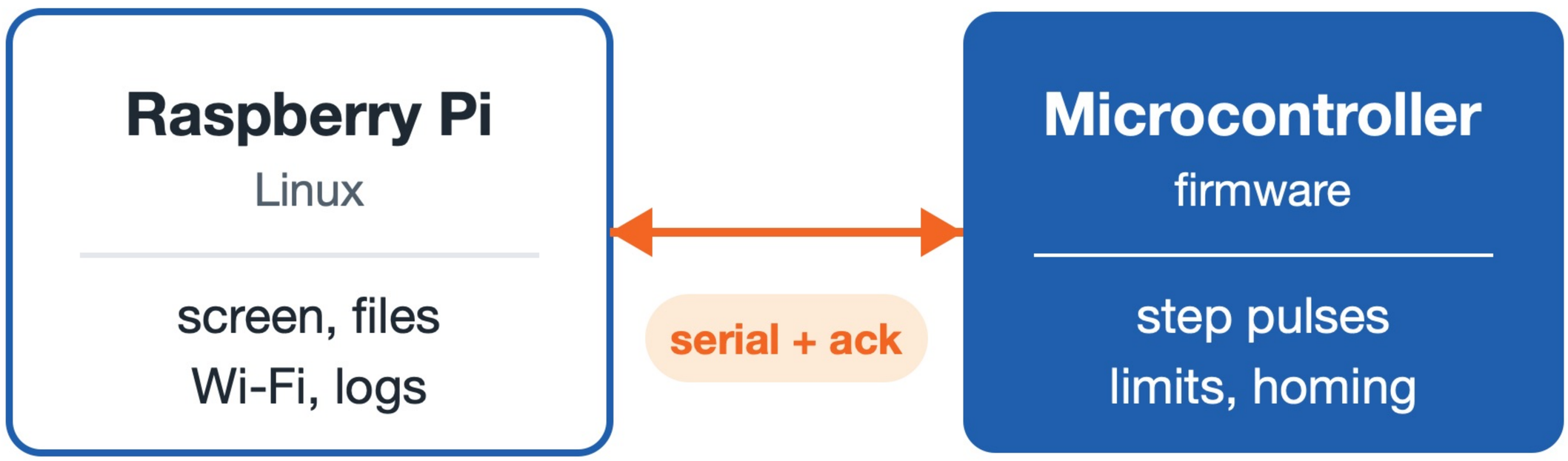
LESSON FROM A CNC MACHINE HMI

Screen on Linux. Timing on an MCU.

How we split the work on an industrial machine controller.



Swipe for the split ->

THE PROJECT

A CNC drill, an operator HMI.

For a European industrial automation company, we built the HMI for a CNC drilling machine. Operators can:

- Pick the machine make and part number
- Set and read 3-axis coordinates
- Upload files and code over Wi-Fi
- Monitor the machine's activity

STACK

HTML/JS screen, Flask, Raspberry Pi

THE TEMPTATION

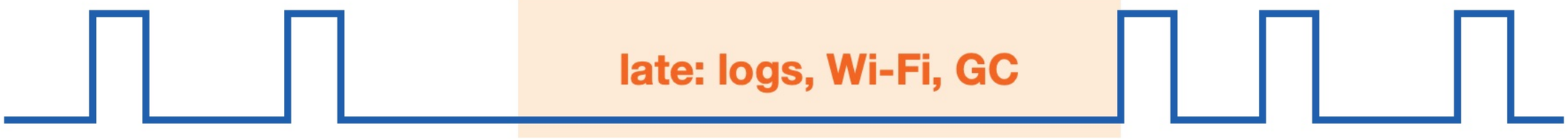
GPIO pins aren't real-time.

The Pi has GPIO, so why not drive the motors from it too? Because standard Linux gives no timing guarantee, and Python adds pauses of its own.

Step pulses you planned



Step pulses you get under load



A late pulse can mean a lost step. A lost step is a hole in the wrong place.

1 ON THE MICROCONTROLLER

Hard timing goes in firmware.

Anything with a deadline the machine can feel runs on the MCU:

- Step and direction pulses
- Limit switches and homing
- Fault handling on each axis
- A watchdog timer

RUNS ON

Bare-metal or FreeRTOS firmware, or a dedicated motion controller.

2 ON LINUX

The Pi does what people touch.

Linux is great at screens, files and networks. Let it do exactly that:

- The operator screen
- Part numbers, coordinates, job files
- Wi-Fi uploads and logs
- Activity monitoring

WHY IT WORKS

A short pause here goes unnoticed.
On a motor axis, it doesn't.

3 THE LINK BETWEEN THEM

Keep the link boring.

A simple serial protocol, with three rules:

- Every command gets an acknowledgement.
- The MCU owns the machine state. The Pi only displays it.

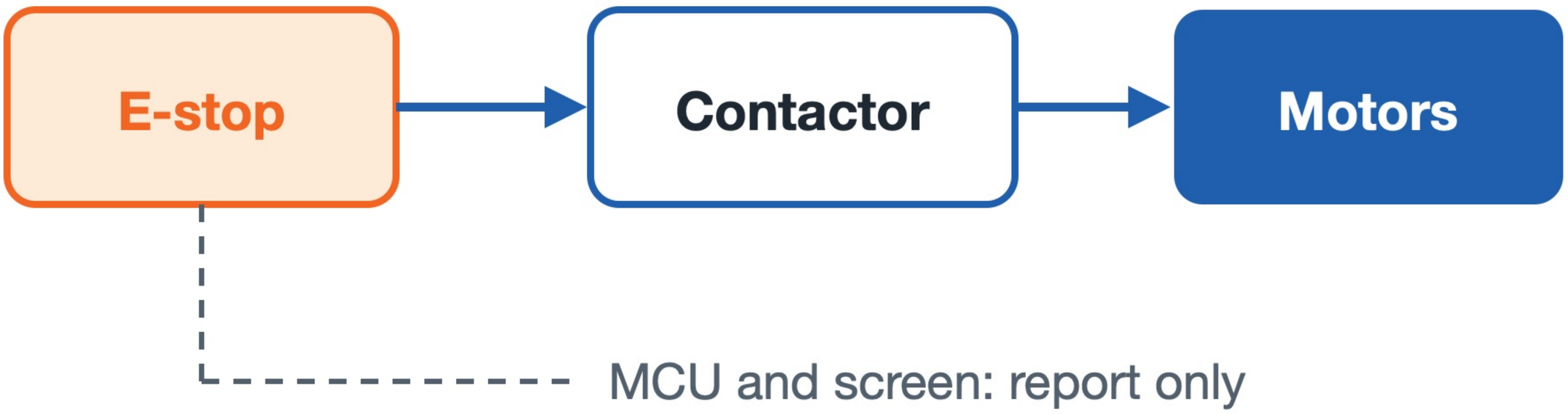
HEARTBEAT

If the Pi goes silent, the MCU stops the axes safely. A frozen screen must never mean a drill that keeps going.

4 EMERGENCY STOP

E-stop is wired, not coded.

The emergency stop cuts motor power in hardware. Firmware sees it and the screen shows it, but neither one is in the path that stops the machine.



Design to your machine's safety standards, e.g. ISO 13850 for the emergency stop function.

THE TAKEAWAY

One job for each processor.

Linux: screen, files, network, logs

MCU: pulses, limits, homing, watchdog

Link: acks and a heartbeat

E-stop: hardware, always

If the screen freezes, the machine should stop safely, not keep drilling.

Follow CoreFragment for more